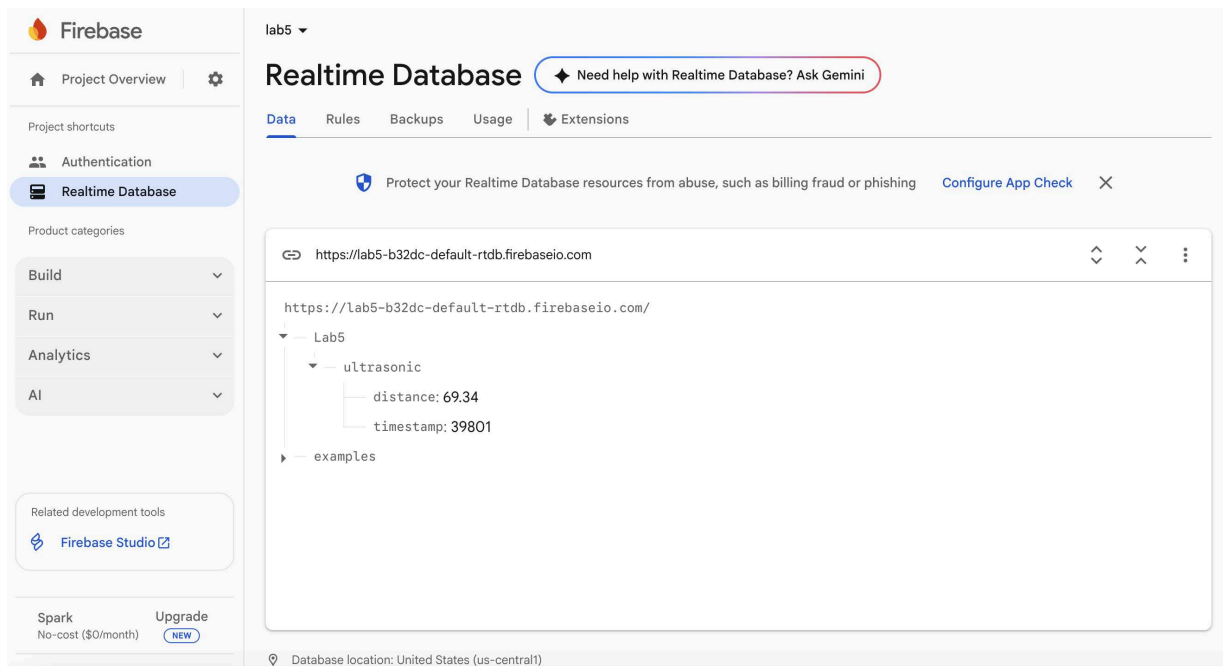


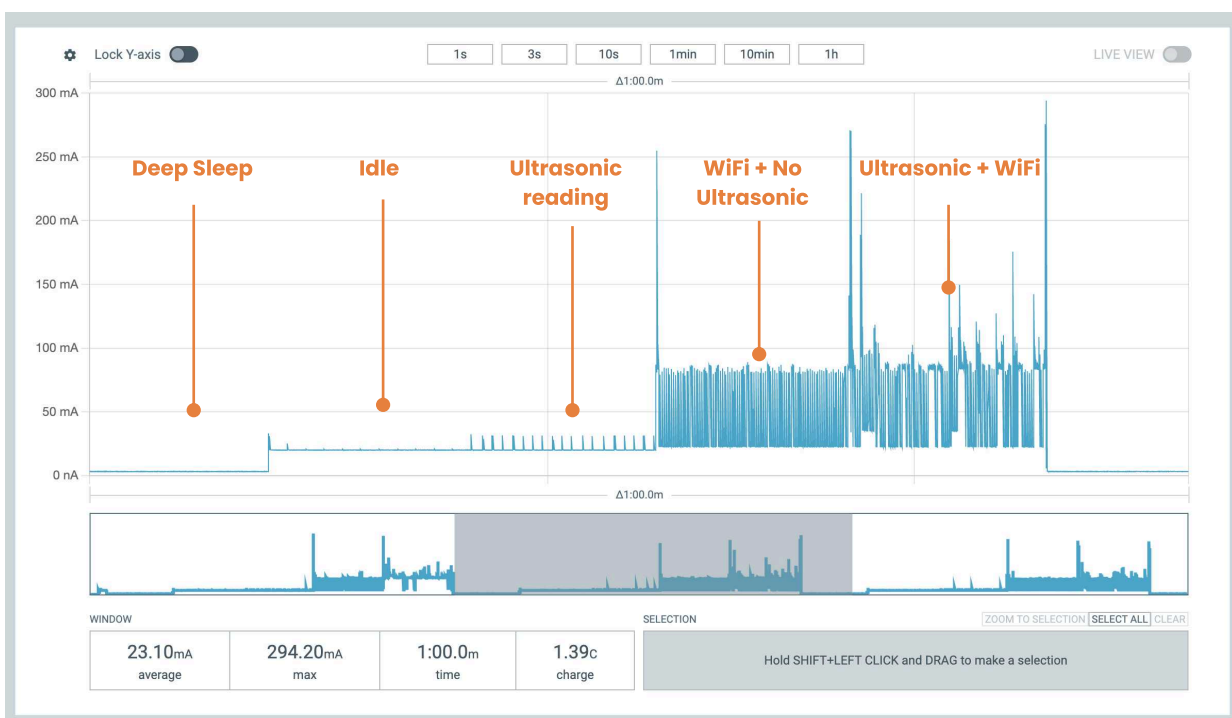
Transmitting your HC-SR04 Data to Firebase

A screenshot on your Firebase RTDB's webpage to show that your data has been successfully uploaded.



Power Consumption Measuring

A screenshot of the plot on your power profiler app to show the power consumption change during a 1 minute time window with 5 different mode or stages. Annotate what each stage stands for.

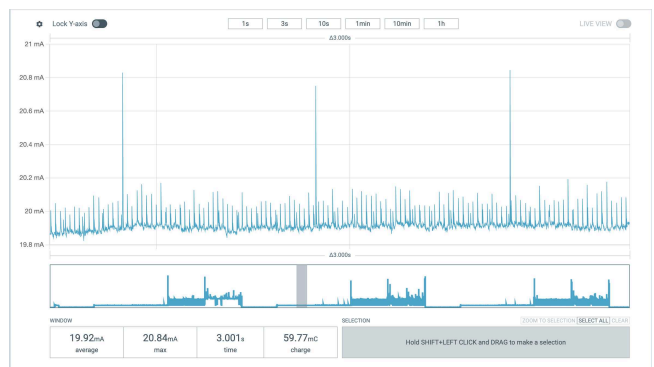


Set the time window from 1 minute to 10 seconds and take a screenshot for each mode/stage. Calculate how long would a 500mAh battery last when the ESP32S3 is in each mode/stage.

Deep Sleep



Idle



Ultrasonic reading



WiFi + No Ultrasonic



Ultrasonic + WiFi



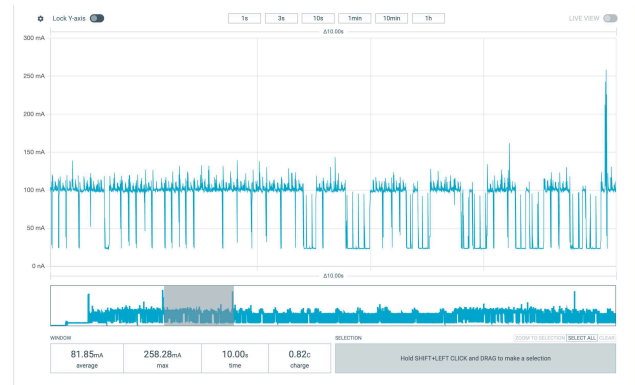
Stage	Avg current (mA)	Battery life (hours)
Deep Sleep	2.99	$500/2.99 = 167.22$ h
Idle	19.92	$500/19.92 = 25.10$ h
Ultrasonic reading	20.18	$500/20.18 = 24.78$ h
WiFi + No Ultrasonic	29.70	$500/29.70 = 16.84$ h
Ultrasonic + WiFi	54.58	$500/54.58 = 9.16$ h

Update your code to change the data transmission rates for WiFi and measure how the power consumption changes. Take a screenshot of the power consumption plots for WiFi. Make plots for WiFi that show the relationship between transmission rates and average current used. Calculate how long a 500mAh battery would last for these new transmission rates.

2 times per second (2 Hz)



1 time per second (1 Hz)



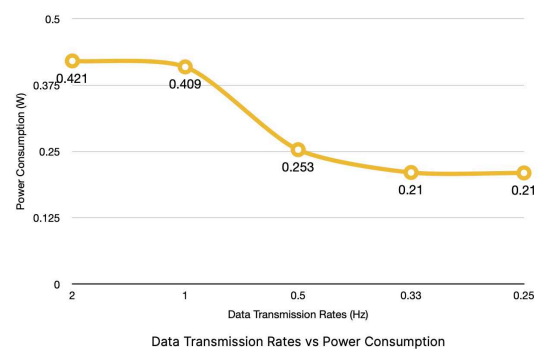
Once every 2 seconds (0.5 Hz)



Once every 3 seconds (0.333 Hz)



Once every 4 seconds (0.25 Hz)

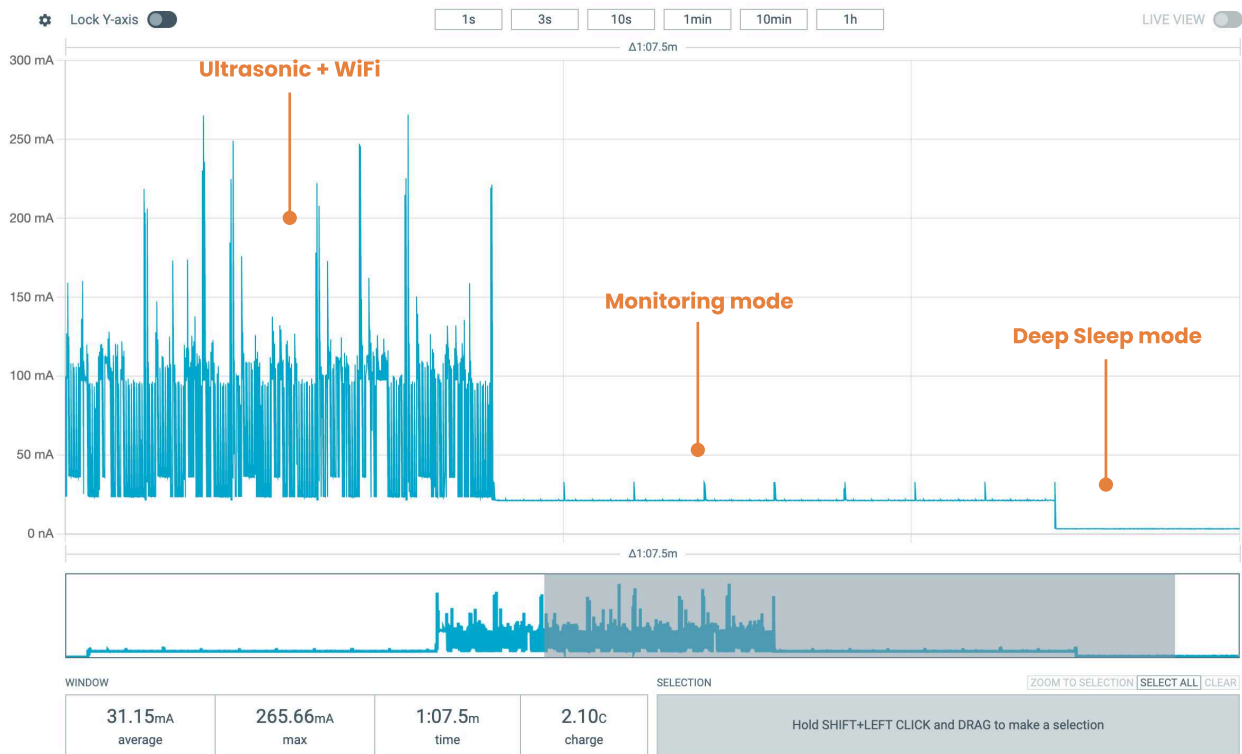


WiFi transmission rate	Avg current (mA)	Power at 5 V (W)	500 mAh life (hours)
2 Hz	84.10	0.4205	5.95 h
1 Hz	81.85	0.4093	6.11 h
0.5 Hz	50.65	0.2533	9.87 h
0.33 Hz	42.05	0.2103	11.89 h
0.25 Hz	41.90	0.2095	11.93 h

Create your own power-saving strategy

My strategy is to keep WiFi turned off most of the time and only enable it briefly when an ultrasonic event is detected and data needs to be uploaded. By throttling upload frequency, immediately disconnecting WiFi after transmission, and using deep sleep when no activity is present, the average current stays low enough for a 500 mAh battery to last at least 24 hours.

Estimation



The power consumption plot shows three main stages. First, during ultrasonic sensing with WiFi enabled, the ESP32 draws the highest current with large spikes caused by WiFi data uploads to Firebase. Next, in monitoring mode, WiFi is turned off and only the ultrasonic sensor is active, resulting in a much lower and steadier current draw with small periodic spikes. Finally, in deep sleep mode, the current drops to near zero, indicating minimal power usage when no activity is detected.

If the device runs continuously with ultrasonic sensing and WiFi enabled, the battery would last only about 9 hours, which does not meet the 24-hour requirement. Using the power-saving strategy shown in this plot reduces the average current to 31.15 mA, extending battery life to about 16 hours. By further increasing time spent in deep sleep and limiting WiFi uploads to brief, event-driven transmissions, the average current can be reduced enough to support 24 hours of operation on a 500 mAh battery.

Github Repo

<https://github.com/yuwennnnnnnnnnnn/514.git>